

Assessing Course Policy Impacts: Late Course Enrollment and Its Effects on Student Performance and Incomplete Grades

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Late Enrollees

- **Late enrollment:** Enrolling after the first lecture
- Presents challenges for completing work on time
- One study indicated that students of color were more likely to enroll late [2]

Incompletes

- **Incomplete:** Grade given to student at risk of failing; allows them to complete outstanding work in a subsequent term



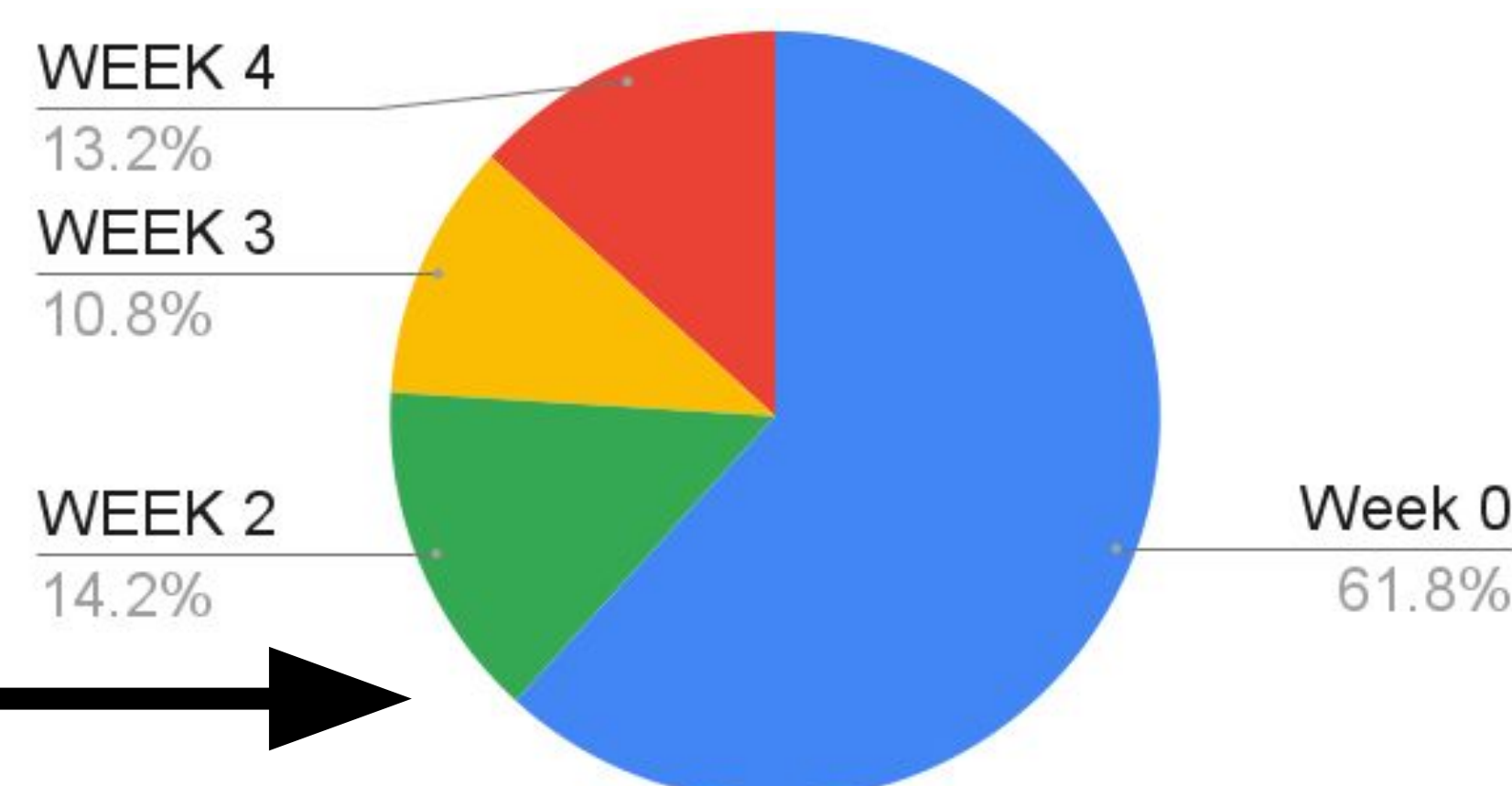
By Week 4, one-third of the semester's lab assignments / exams are due

RQ: How does late enrollment affect students' likelihood of receiving Incomplete grades and their overall academic performance compared to on-time enrollees?

Data Collection

- Fall 2023 dataset of CS0 at UC Berkeley
- 91% participated in the survey
- Students given extended deadlines with no penalty in CS0

Number of students enrolled by week



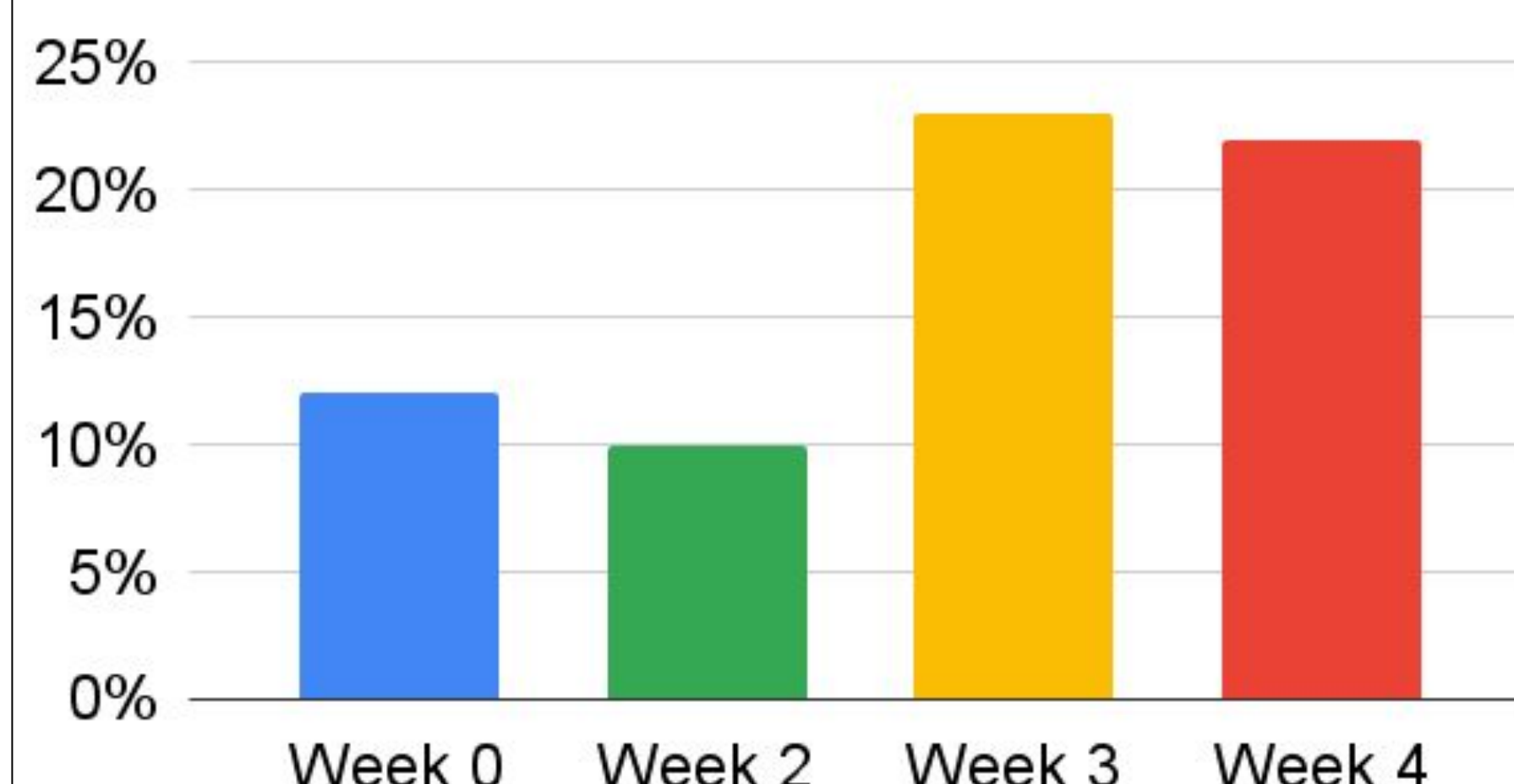
Results

Late enrollment is associated with significantly **lower performance** and **higher Incomplete rates**, for students joining after Week 2

Percentage that Used OH by Enrollment Week



Percentage of CS/EECS Majors by Week

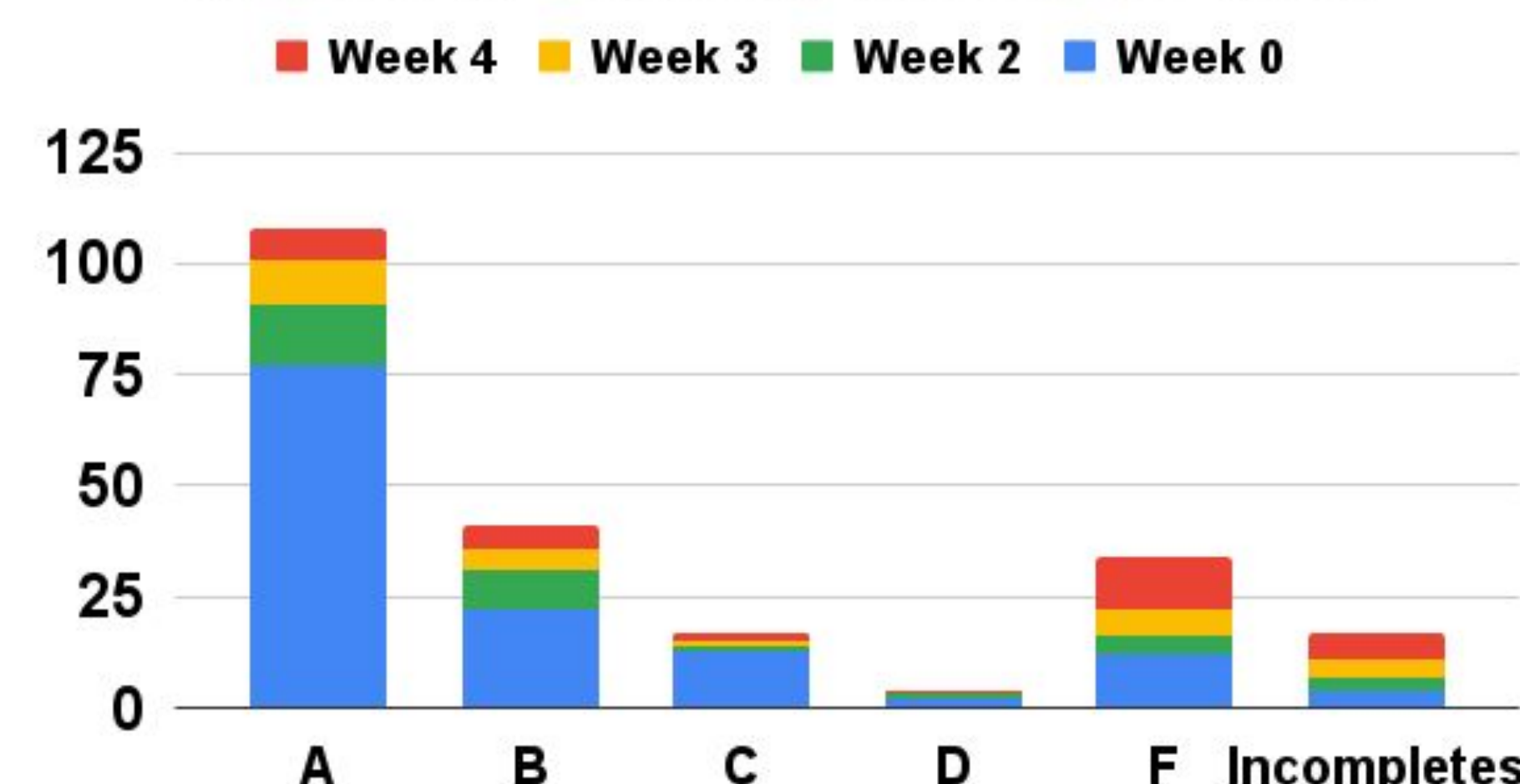


- Later enrolling students used course resources less.

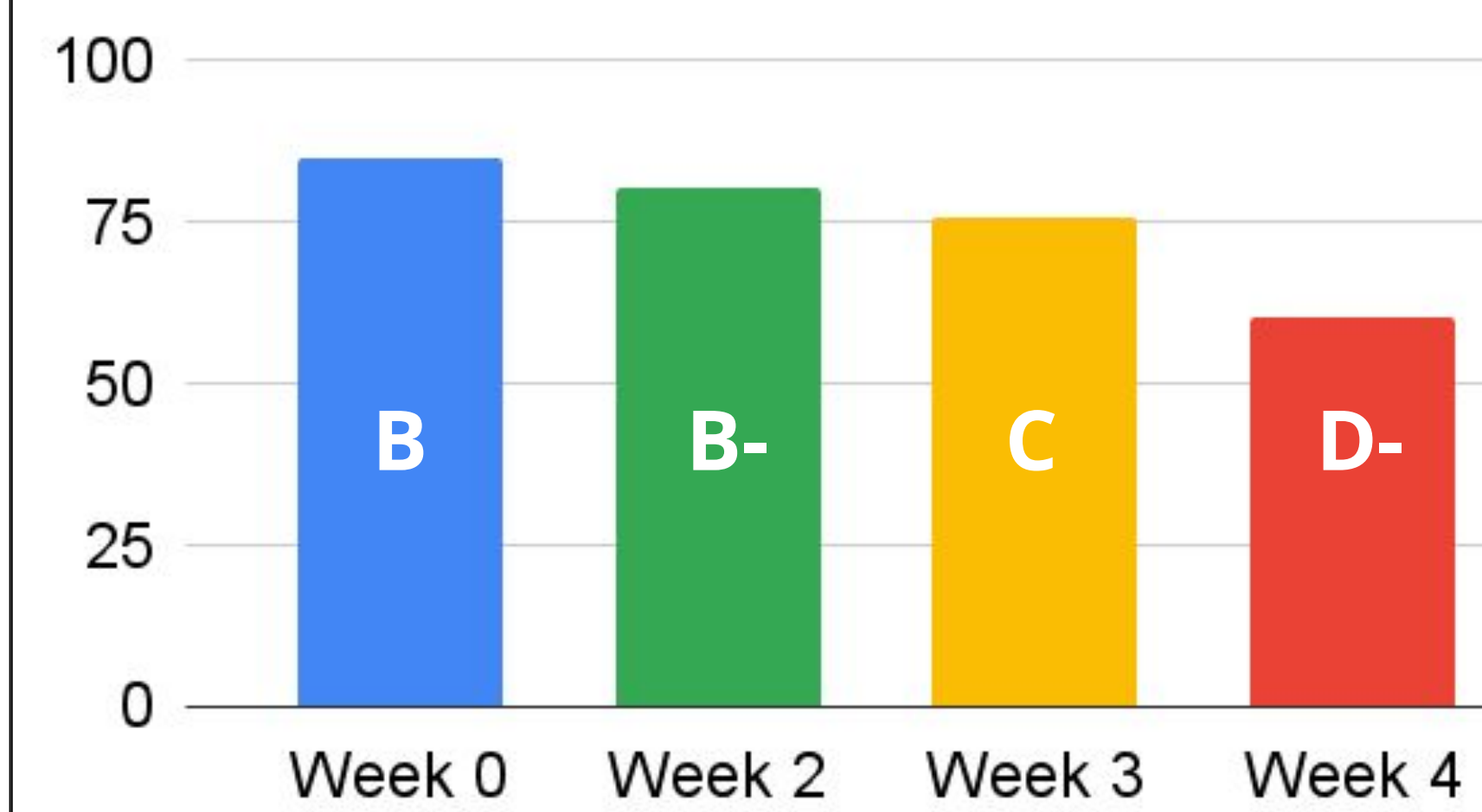
- Later enrollees were likelier CS/EECS majors usually after dropping CS1

Results Continued

Number of Grades by Enrollment Week

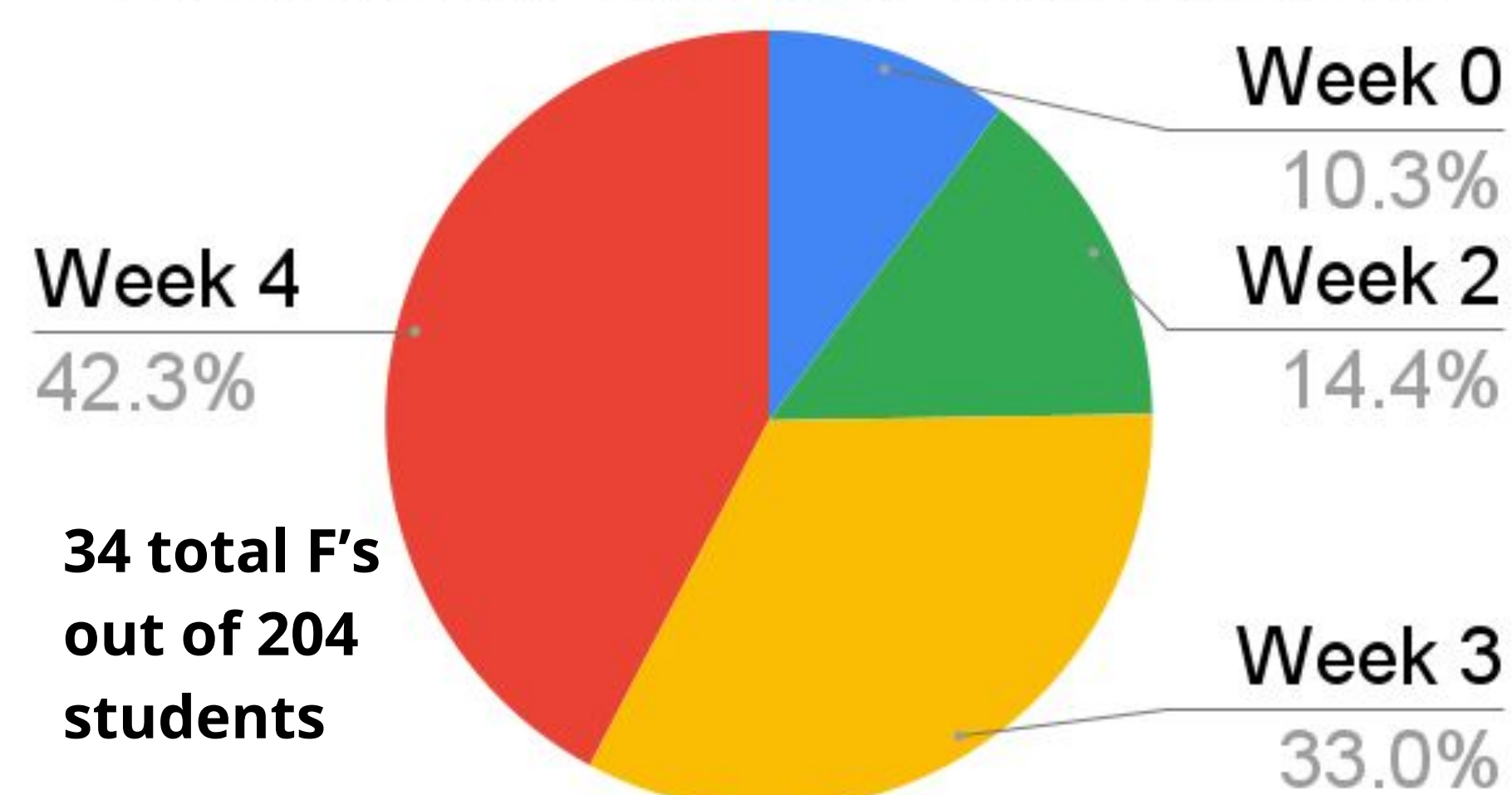


Percentage of Points Earned by Enrollment



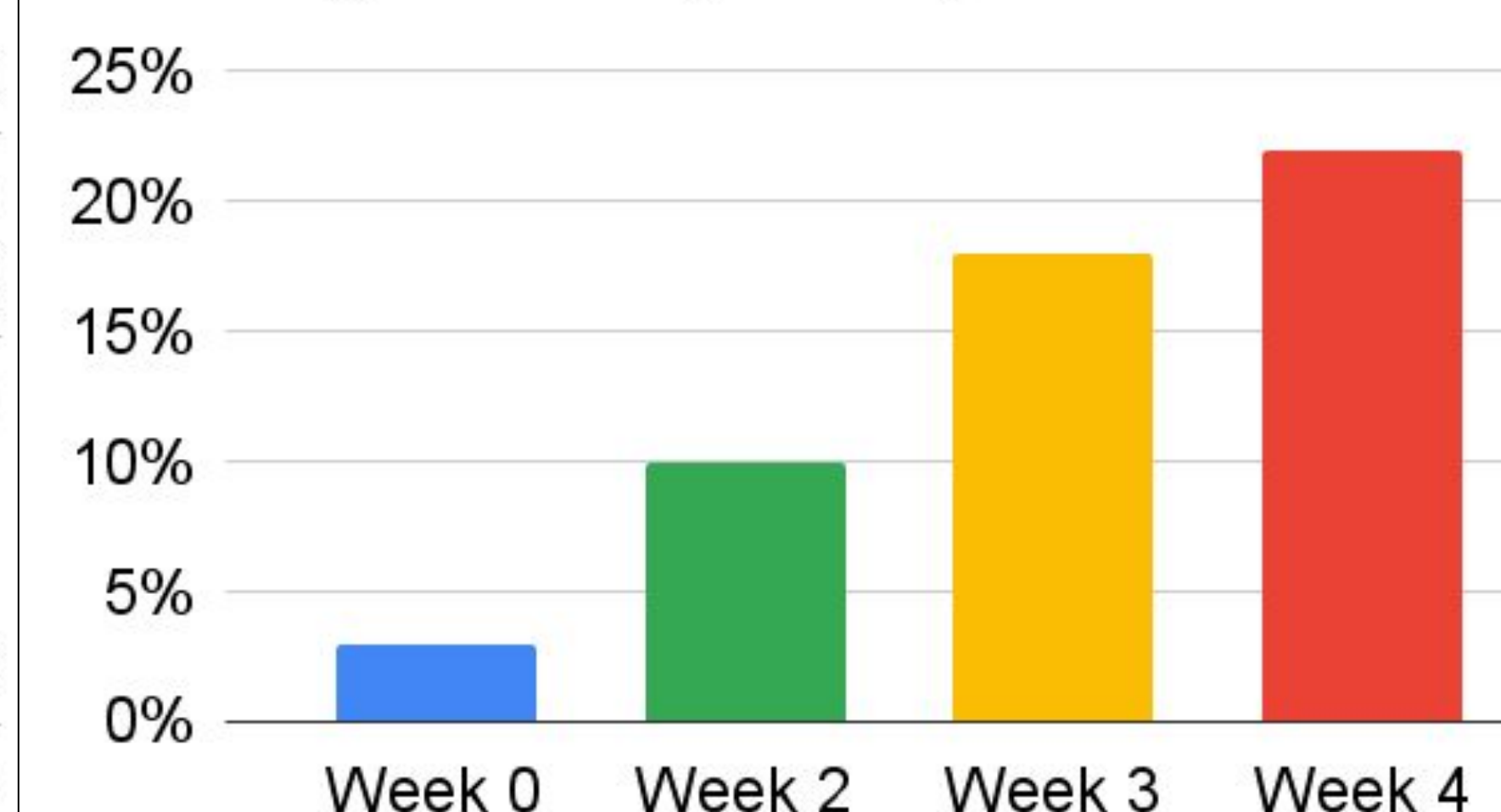
The average grade for on-time students was a B, while late enrollee averages were B- for Week 2, C for Week 3, and D- for Week 4.

Percentage of F Grades by Enrollment Week



- Students were more likely to receive an F grade the later they enrolled.

Percentage of Incompletes by Enrollment Week



- More likely to receive an Incomplete the later they enrolled.

Future Work

1. Explore the motivations behind late enrollment - qualitative interviews
2. Examine key factors to identify barriers faced by late enrollees
 - a. Inform tailored support strategies
3. Collect data over multiple terms
 - a. Observe trends, assess the effectiveness of policy interventions
4. Share this data with CS1 students on day 1 to nudge to CS0 earlier (week 1).
5. Consider a "CS1 placement test" to nudge to CS0 earlier (week 0).

References

- [1] Ben Littlepage. 2015. Influence of Academic Goal Commitment on Late Registrant Success. *Journal of College Orientation, Transition, and Retention* 23, 1 (Dec. 2015). <https://doi.org/10.24926/jcotr.v23i1.2895>
- [2] Kathy Maalouf. 2012. The Influence of Late Registration on Academic Outcomes and Retention at a Multi-Campus Community College. Doctoral dissertation. Old Dominion University.
- [3] Letizia Jaccheri Michail N. Giannakos, Ilias O. Pappas and Demetrios G. Sampson. 2016. Understanding student retention in computer science education: The role of environment, gains, barriers and usefulness. *Education and Information Technologies* 22 (2016), 2365–2382